MIL-E-1/1012A(EL)
25 May 1964
SUPERSEDING
MIL-E-1/1012(SigC)
14 February 1956

MILITARY SPECIFICATION

ELECTRON TUBE, KLYSTRON
TYPES SRL-7F, SRL-7G, SRL-7H

The complete requirements for procuring the electron tube(s) described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Reflex, Integral Cavity, Mechanically Tuned: SRL-7F=1700 to 1930 Mc (Note 1)
SRL-7G=1930 to 2160 Mc (Note 1)
SRL-7H=2160 to 2400 Mc (Note 1)

RATINGS:

| Parameter: | Ef | Ers | Irs | Eg | Pl | Er | T | Alt |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | mAdc | Vdc | W | Vdc | °C | ft |
| Maximum: | 6.9 | 1000 | 220 | +50 | 220 | -1500 | (Note 3) | 10,000 |
| Minimum: | 5.7 | --- | --- | --- | --- | --- | --- | --- |

| TEST CONDITIONS: | 6.3 | 1000 (Note4) | --- | Connect to k | --- | -350 to -950 | --- | --- |
|---|---|---|---|---|---|---|---|---|

OUTLINE AND DIMENSIONS: See Fig. 1 MOUNTING POSITION: Any

BASE: Shortened (See Fig. 1) Medium Shell Octal B8-11, Low Loss Phenolic
COOLING PRECAUTION: (Note 3)
CATHODE: Coated Unipotential

| PIN NO.: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Tube Shell | Top Cap |
|---|---|---|---|---|---|---|---|---|---|---|
| ELEMENT: | lc | h | ic | lc | g | lc | h-k | h-k | m | r |

PIN NO.: Side Terminals: Type N (Mates with R-F Plug UG-21B/U or equivalent)
ELEMENT: RF Output

| MIL-E-1 Ref. Par. | Test | Conditions | AQL (percent defective) | Insp. Level or Code | Symbol | Limits Min. | Limits Max. | Unit |
|---|---|---|---|---|---|---|---|---|
| | GENERAL | | | | | | | |
| --- | Preproduction Sample Approval | (Note 5) | --- | --- | --- | --- | --- | --- |
| 3.6 | Performance | (Note 6) | --- | --- | --- | --- | --- | --- |
| 3.7 | Marking | (Note 7) | --- | --- | --- | --- | --- | --- |
| 4.5 | Holding period | t=168 hrs. (Note 8) | --- | --- | --- | --- | --- | --- |
| 4.6 | Preheating | (Note 4) | --- | --- | --- | --- | --- | --- |
| 4.9.2 | Dimensions | (Note 9) | --- | --- | --- | --- | --- | --- |



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| MIL-E-1 Ref. Par. | Test | Conditions | AQL (percent defective) | Inspection level or code | Symbol | Limits Min. | Limits Max. | Unit |
|---|---|---|---|---|---|---|---|---|
| | **ACCEPTANCE INSPECTION, PART 1 (PRODUCTION) - (Note 10)** | | | | | | | |
| 4.10.8 | Heater current | (Note 11) | 0.65 | II | $I_f$ | 1.8 | 2.2 | A |
| 4.10.4.8 | Resonator current | $E_r$ = -350 to -950 Vdc<br>$E_g$ = 0<br>(Note 12) | 0.65 | II | $I_{rs}$ | --- | 0.22 | Adc |
| --- | Emission | (Notes 12,13)<br>t = 60 sec. | 0.65 | II | $\Delta I_{rs}$ | --- | 12 | % |
| 4.10.5.4 | Reflector voltage:<br>F1<br>F3 | Power output(1) | 0.65 | II | <br>$-E_r$<br>$-E_r$ | <br>350<br>--- | <br>---<br>950 | <br>Vdc<br>Vdc |
| 4.10.6.7.1 | Total reflector current | Er = -300 to -1000<br>(Notes 8,12,14) | 0.65 | II | $I_r$ | --- | 10 | uAdc |
| 4.10.6.7.1 | Total reflector current | Power output(1) | 0.65 | II | $I_r$ | --- | 10 | uAdc |
| 4.10.7.3.2 | Tunable frequency | Power output(3)<br>(Notes 2,15) | 0.65 | II | F | F1 | F3 | Mc |
| 4.15.1 | Power output(1) | Er =-350 to -950 Vdc<br>Eg = 0<br>LR = 50 ± 5 ohms<br>L(VSWR) = 1.1:1 max<br>F1, F3 only<br>(Notes 2,16) | 0.65 | II | $P_O$ | 5.0<br>(Note 16) | --- | W |



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| MIL-E-1 Ref. Par | Test | Conditions | AQL (percent defective) | Inspection level or code | Symbol | Limits Min. | Limits Max. | Unit |
|---|---|---|---|---|---|---|---|---|
| | | ACCEPTANCE INSPECTION, PART 1 (PRODUCTION) - (Note 10) - (Continued) | | | | | | |
| --- | Reflector-voltage modulation sensitivity | Power output(1) (Note 17) | 0.65 | II | $\Delta F/\Delta E_r$ | 55 | 130 | Kc/V |
| --- | Static frequency linearity | Power output(1) (Note 18) | 0.65 | II | --- | --- | 20 | % |
| | | ACCEPTANCE INSPECTION, PART 2 (DESIGN) | | | | | | |
| 4.15.1 | Power output(2) | Power output(1), except F2 only | 6.5 | L6 | $P_O$ | 5.0 (Note 16) | --- | W |
| --- | Static frequency linearity | Power output(3) (Note 18) | 6.5 | L6 | --- | -- | 20 | % |
| --- | Carrier-frequency Shift | Power output(1) (Note 19) | 6.5 | L6 | $\Delta F$ | --- | ±2 | Mc |
| 4.9.19.1 | Low-frequency vibration | No voltages<br>G = 2.3<br>F = 25 ± 2 cps<br>t = 120 sec | 6.5 | L6 | --- | --- | --- | --- |
| | | ACCEPTANCE INSPECTION, PART 3 (PERIODIC) - (Note 20) | | | | | | |
| 4.15.1 | Power output(3) | Power output(1), except continuous sweep over entire range F1 through F3 | --- | --- | $P_O$ | 5.0 | --- | W |
| --- | Reflector-voltage modulation sensitivity | Power output(3) (Note 17) | --- | --- | $\Delta F/\Delta Er$ | 55 | 130 | kc/V |



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| MIL-E-1 Ref. Par. | Test | Conditions | AOL (percent defective) | Inspection level or code | Symbol | Limits Min. | Limits Max. | Unit |
|---|---|---|---|---|---|---|---|---|
| | ACCEPTANCE INSPECTION, PART 3 (PERIODIC) - (Note 20) - (Continued) | | | | | | | |
| --- | Amplitude modulation | Power output(3) Deviation = ± 4.5 Mc | --- | --- | -- | -- | 30 | % |
| --- | Noise | Power output (3) | --- | --- | AM | --- | 3 | % |
| | | | | | Random FM | --- | ± 9 | kc |
| | | | | | AC filament hum | --- | ± 9 | kc |
| --- | Carrier-frequency shift | Power output (3) (Note 19) | --- | --- | ΔF | --- | ±2 | Mc |
| 4.10.14 | Direct inter-electrode capacitance | No voltages Reflector-to-shell | --- | --- | C | -- | 6 | pf |
| 4.15.5 | Temperature coefficient | Power output(3) T =-40° to + 55°C (Note 21) | --- | --- | ΔF/ΔT | --- | 50 | kc/°C |
| -- | Mechanical-tuning sensitivity | Power output(3) each tube type ± 1/2-turn of tuner screw | | | ΔF | 10 | 50 | Mc |
| 4.9.17 | Mechanical fatigue(1) | Trimmer screw, each tube type ± 1/8-turn of trimmer screw | --- | --- | --- | 10,000 | --- | cycles |
| 4.9.17 | Mechanical fatigue(2) | Tuner range F1 through F3, each tube type | --- | --- | --- | 50 | --- | cycles |

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| MIL-E-1 Ref. Par. | Test | Conditions | AQL (percent defective) | Inspection level or code | Symbol | Limits Min. | Limits Max. | Unit |
|---|---|---|---|---|---|---|---|---|
| | ACCEPTANCE INSPECTION, PART 3 (PERIODIC) - (Note 20) - (Continued) | | | | | | | |
| --- | Temperature-humidity operation | Power output (2) (Note 22) | --- | --- | $P_O$ | 5.0 | --- | W |
| | ACCEPTANCE INSPECTION, PART 3 (LIFE) | | | | | | | |
| | Group D | | | | | | | |
| 4.11 | Life test | Er = -900 Vdc non-oscillating | --- | --- | t | 1000 | --- | hrs |
| 4.11.4 | Life-test end points | Power output(1) (Note 23) | --- | --- | $P_O$<br>Carrier shift<br>$\Delta F$ | 4.0<br>---<br>--- | ---<br>---<br>5.0 | W<br>---<br>Mc |
| 4.9.18<br>5. | Container drop<br>Preparation for delivery | Required<br>(Note 24) | --- | --- | --- | --- | --- | --- |
| | | | --- | --- | --- | --- | --- | --- |



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NOTES:

1. Although the successive frequency ranges are contiguous, they serve to discriminate the respective Klystron types covered herein. (See Note 2, below.)

2. For test purposes, each frequency range is defined and subdivided into "F1", "F2", "F3" frequency categories as follows:

| | F1 | F2 | F3 |
|---|---|---|---|
| SRL-7F | 1700 ± 0.3% Mc | 1815 ± 0.3% Mc | 1930 ± 0.3% Mc |
| SRL-7G | 1930 ± 0.3% Mc | 2045 ± 0.3% Mc | 2160 ± 0.3% Mc |
| SRL-7H | 2160 ± 0.3% Mc | 2280 ± 0.3% Mc | 2400 ± 0.3% Mc |

3. Air cooling of at least 50 CFM is necessary at maximum power input at sea level. Nevertheless, sufficient air cooling should be provided to keep the envelope temperature below +100°C. The temperature of the base and of the reflector electrode insulation should not exceed +75°C.

4. Full heater voltage of 6.3V shall be applied for 2 minutes and stabilization of heater current obtained before any other voltages are applied.

5. The Preproduction Sample Approval requirements in Electronics Command Drawing SC-A-46600 hereby replace any Qualification requirements referable to the product(s) covered herein. (Copies of Electronics Command Drawing SC-A-46600, "Preproduction Sample Approval In Lieu of Qualification Requirements In Specifications" may be obtained from the procuring activity or as directed by the contracting officer.)

6. Associable with and in addition to the performance requirements specified directly herein, the following tests and requirements listed in Specification MIL-E-1, paragraph 3.6, shall be applicable also: (paragraph) 3.3, 3.3.1, 3.4.1, 3.4.3, 3.8.4.1, 4.3, 4.4, 4.9.1, 4.9.4, 4.9.5, 4.9.5.1, and 4.9.21.

7. Type-designation marking for the Klystron(s) covered herein shall be as required in Drawing SC-A-46600. (See Note 5 above.)

8. Holding-period requirement is applicable particularly to the Total Reflector Current test in Measurement Acceptance Tests, Part 1, herein.

9. All dimensions shall be in accordance with Figure 1 herein. Inspection for dimensions shall be in accordance with the applicable AQL and Inspection Level data presented in the table on Figure 1.



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NOTES - (Cont'd):

10. The AQL for the combined defectives for attributes in Acceptance Inspection, Part 1, herein, excluding Mechanical, shall be 1.0 %. A tube having one or more defects shall be counted as one defective.

11. (Refer to Note 4, above.) The heater current shall stabilize to the limit(s) specified, within 2 minutes.

12. Tube shall not be oscillating during this test.

13. Test shall be performed under Ers and Eg conditions specified herein for Resonator Current test, and the filament voltage is decreased from 6.3V to 5.7V. After 1 minute, measurement(s) shall be made in accordance with the following formula:

$$\frac{I_{rs}\ (\text{at } E_f = 6.3V) - I_{rs}\ (\text{at } E_f = 5.7V)}{I_{rs}\ (\text{at } E_f = 6.3)} \times 100$$

The computed percentage shall not exceed the maximum value specified, and shall be interpreted as the Emission (characteristic) for the tube.

14. No connections are to be made to pins marked "ic" (internal connection). For Eg=0, connect grid to cathode. Apply heater voltage per Note 4 above. Apply other applicable voltages for one minute, and then read total reflector current.

15. Tuning of the tube over the applicable frequency range shall be in accordance with the following maximum turning requirements for each respective tuning screw:

SRL-7F: 1-3/4 ± 1/4 turns
SRL-7G: 1-1/2 ± 1/4 turns
SRL-7H: 1-1/8 ± 1/4 turns

To tune, rotate each tuning screw not more than 1/2 turn at a time: clockwise - to increase frequency, counterclockwise - to decrease frequency.

16. The specified minimum power output, except as indicated in (c) below, shall apply in accordance with the following:

(a) Under the conditions specified, at the basic frequency value(s) indicated in the tabulation in Note 2, above.

(b) For any reflector voltage between the values required to produce a frequency change of ± 4.5 Mc from the carrier frequency. (See exception in (c), below.) Carrier frequency is defined as that point where the reflector voltage is adjusted for minimum deviation sensitivity.

(c) For tube SRL-7F: at reflector voltage producing a frequency change of −4.5 Mc from carrier frequency, the power output shall be 4.0 W minimum; however, the power output shall be 5.0 W minimum when the tube is tuned to 1715 Mc −4.5 Mc instead of 1700 Mc −4.5Mc.



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NOTES -(Cont'd):

17. The reflector-voltage modulation sensitivity shall be determined at the operating points used for the power output measurements.

18. Deviation from linearity shall not exceed the value specified for the ±4.5 Mc frequency deviation, and shall be within the limit specified herein relative to the test definition and formula given below. There shall be only one inflection point in the reflector-voltage frequency curve within the ±4.5 Mc deviation band. "Static frequency linearity" is derivable from the following computation:

$$100 - \frac{(\text{Small-signal modulation sensitivity})(\Delta E_r \text{ for } \pm 4.5 \text{ Mc change})}{9000 \text{ kc}} \times 100 \ \%$$

19. The change in the Klystron center frequency (carrier), when the frequency of the modulating voltage is varied from 60 cps to 5 Mc and the modulation about the carrier is varied from 0 to ±4.5 Mc, shall not exceed the amount specified.

20. For this test group, "Part 3 (Periodic)", the requirements in Specification MIL-E-1, paragraph 4.1.1.2.2 shall apply except that report of any tube failures shall be directed to Procurement and Production Directorate, Fort Monmouth, New Jersey, Attention: AMSEL-PP-E CMD.

21. The tube shall be operated at the frequencies and with applied voltages as applicable per the specified power-output test conditions. At the initial frequency, the tube shall be allowed to stabilize during operation at constant ambient temperature (initial temperature point) for at least 10 minutes. The specified temperature-coefficient limit shall not be exceeded within any respective 30°-interval throughout the temperature range covered.

22. The Klystron shall be capable of specified operation under full load and under each of the following environmental conditions:

    (a) At 90% relative humidity, and at $T_A$=+40°C, for 96 hours.

    (b) At $T_A$=+55°C for 4 hours.

    (c) At $T_A$=-40°C for 4 hours.

23. For each tube subjected to life test, Life Test shall be considered terminated when the output power falls below 4.0W, when the carrier-frequency shift with modulation (see Note 19 above) exceeds ±5.0 Mc, or when the tube fails to meet any other Acceptance Inspection Part 1 test herein.



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NOTES - (Cont'd):

24. Tubes shall be prepared for domestic or overseas shipment as specified in the contract or order.

25. Tubes subjected to the following destructive tests are not to be accepted on the contract or order:

Mechanical fatigue (1)
Mechanical fatigue (2)
Temperature-humidity operation
Life test

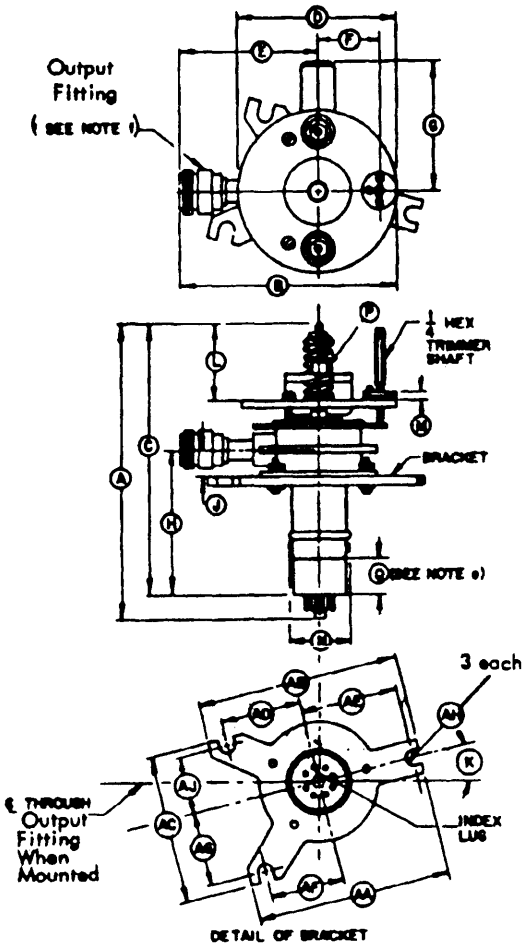
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| SYM. | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL | LIMITS | |
|---|---|---|---|---|
| | | | MIN. | MAX. |
| | QUALIFICATION TESTS | | | |
| P | CAP C1-1 (SEE NOTE d) | | | |
| Q | --- | --- | .765 | .797 |
| | MEASUREMENTS ACCEPTANCE TESTS, PART 1 | | | |
| A | | | --- | 1-5/8 |
| B | | | --- | 1-7/8 |
| C | | | --- | 1-7/16 |
| D | | | --- | 1-7/8 |
| E | | | --- | 1-7/8 |
| G | (SEE NOTE e) | I | --- | 1-7/8 |
| H | | | --- | 1-7/8 |
| J | | | 11/16 | 13/16 |
| L | | | --- | 1-7/8 |
| M | | | --- | 1-7/8 |
| | NOMINAL DIMENSIONS | | | |
| N | 1-7/8 dia | | | |
| P | 1-25/64 | | | |
| R | .750 | | | |
| S | 1-17/32 dia | | | |
| AA | 1-7/16 | | | |
| AB | 1-7/8 | | | |
| AC | 1-7/8 | | | |
| AD | 1-7/8 | | | |
| AE | 1-7/8 | | | |
| AF | 1-7/16 | | | |
| AG | 1-7/16 | | | |
| AH | 1-7/16 | | | |
| AJ | 1-7/16 | | | |
| AK | 1-7/8 | | | |

NOTES:

a. All dimensions in inches, unless otherwise specified.

b. Nominal dimensions are for information and are not required for inspection purposes.

c. The AQL for the combined mechanical defectives in measurements acceptance tests, part 1, shall be 1 percent.

d. Refer to Specification MIL-E-1.

e. Base is medium shell octal, B8-11, as shown in MIL-E-1 except for dimension Q. Base dimensions to be inspected on qualification basis.

f. Type N fitting. Mates with connector type UG-21B/U, or equivalent.

Figue 1. Outline and dimensions.



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

| SPECIFICATION | | |
|---|---|---|
| ORGANIZATION | CITY AND STATE | |
| CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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